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Nanoindentation-enhanced screening of hydraulic fracturing fluid additives

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ABSTRACT

This paper describes a viable nanoindentation-enhanced technique for screening hydraulic fracturing fluid additives via quantifying the effects of rock-fluid interactions on the mechanical properties of a clay-rich shale. Big data nanoindentation testing was conducted on both the intact shale and counterparts hydrothermally treated by four fracturing fluids, including water, KCl solution, and two surfactants, to extract simultaneously the statistical, cross-scale mechanical properties of both the bulk rock and individual constituent phases (e.g., clay matrix, carbonate, and quartz), accompanied by scanning electron microscopy for surface imaging and elemental mapping. Results show that pure water and KCl solution significantly degrade the clay matrix's Young's modulus and hardness by up to 39.4–42.5% and 42.9–57.1%, respectively, which can be mitigated or prevented by appropriate surfactant additives. The physicochemical rock-fluid interactions, such as dissolution of carbonate, are mainly responsible for the property degradation, as evidenced by the hydrothermal treatment-induced large cavities and secondary pores on the surface that tend to impair microstructural integrity of the shale. The big data nanoindentation technique can be implemented as an adaptive toolkit for screening, tailoring, and optimizing the chemical additives in fracturing fluids used for a specific shale formation to prevent shale softening, protect reservoir stability, and maximize well production.

1. Introduction

Shale is a kind of fine-grained clastic rocks constituted from the burial, compaction, and diagenesis of argillaceous sediments. In terms of its composition, highly compacted tiny clay particles of <2 μm, usually along with interparticle nanopores and organic matter (e.g., kerogen), form a macroscopically homogeneous and continuous matrix, in which a variety of other coarse-grained minerals, being either primary or secondary, such as quartz, feldspar, pyrite, and calcite, are embedded as floating, isolated solid inclusions (Katahara, 2008). Neoformation of some secondary phases as precipitates results in pore infills and interparticle cementation (Luo et al., 2021), leading to enhancement in strength and further reduction in the overall porosity and individual pore sizes. Such a microstructure leads to an extremely low permeability (i.e., in the nanodarcy range) of shale formations (Neuzil, 2019; Zendehboudi et al., 2017; Zhao et al., 2018), usually six orders of magnitude lower than conventional sandstone reservoirs with permeability of tens to hundreds of millidarcies. Therefore, traditional hydrocarbon extraction and production methods usually fail to work in such unconventional reservoirs as shales due to their extremely low permeability (Salama et al., 2017).

In current practices, horizontal drilling and hydraulic fracturing, assisted by other stimulation techniques, are used to recover oil and gas from shale reservoirs and other tight formations. Aqueous fracturing fluid, typically consisting of 99% water and proppants and 1% chemical additives (e.g., clay stabilizer and acid), is injected under a high pressure into the target tight formation to create extensive fracture networks and to promote the flow of oil and gas (Norris et al., 2016). However, a more recently recognized issue is that the average output of shale wells reduces rapidly as they start to age after the initial 8–12 months of production (Baihly et al., 2010; Valkó and Lee, 2010). Sufficient data obtained from past and existing operations suggest that the rapid reduction in well production rate may result from shale softening and pertinent proppant embedment, which are largely attributed to the

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physical interactions and chemical reactions between the constituent minerals (e.g., clay minerals, carbonate, pyrite) of rock formations and hydraulic fracturing fluids during and after the fracturing stimulation. For instance, the clay minerals in fine-grained shales can interact with water via such processes as hydration, diffusion, absorption, and adsorption, among others. The negative impacts of these physicochemical interactions usually include swelling and softening of the clay matrix and hence the bulk rock, leading to significant proppant embedment and closure of pre-induced cracks (Chenevert, 1970). Moreover, laboratory data show that the average embedment depth of proppants into the fracture surface can be as large as half of the proppant size (e.g., ~140 μm) in the presence of water flow, which causes an 88% permeability loss in the fracture network (Zhang et al., 2015a, 2015b).

Generally, a suite of specially designed chemical additives is used by the oil and gas industry, usually at a very low concentration (e.g., <1%), to induce important changes in the properties of fracturing fluids and hence to promote oil/gas recovery, maintain wellbore stability, minimize damages to the formation, and facilitate flowback of the fluid (Armstrong et al., 1995; Ferrer and Thurman, 2015). For example, potassium chloride (KCl) is a popular inhibitor used in the water-based fracturing fluids to dehydrate and stabilize the young, active shale formations that contain swelling clay minerals (e.g., smectites). It has been widely accepted that such a treatment is effective because the K^+ cations added to the fluid can exchange the interlayer Na^+ or Ca^{2+} cations in smectites and then form a one-layer water molecule hydrates in the interlayer space, resulting in a reduction in the basal spacing of smectite from 15.0 to 12.5 Å (Wilson and Wilson, 2014). In addition, by using high-salinity fluids, it is possible to stimulate osmotic backflow of the porewater from the rock towards the wellbore, which offsets the hydraulic inflow of mud filtrate (van Oort, 1997). Moreover, the combination of KCl and polymer is also considered in practices to further address the issue of water imbibition and improve the water recovery rate during multi-staged fracturing (Rabie et al., 2019a, 2019b; Yang et al., 2017), because the polymer can encapsulate the fracture surface to prevent the shale-fluid interactions and hence amelioration of the formation wettability (Xu et al., 2018). Also, hydrochloric acid serving as a scale control agent is commonly used to clean the well perforations (Ghalambor and Economides, 2000), but the dissolution of carbonate cementation due to acidizing can also significantly reduce the mechanical properties of rock formations.

As discussed above, understanding how the mechanical properties of shales change upon interactions with fracturing fluids plays a vital role in suppressing shale softening and maximizing the long-term production of hydrocarbons from the formations. The macroscopic softening behavior of shales has been studied by conducting traditional large-scale mechanical measurements (e.g., triaxial testing and pulse transmission technique) on rock core samples (Lin and Lai, 2013; Valès et al., 2004; Wong, 1998). The elastic properties and compressive strengths show clear disparities between the intact samples and those exposed to diverse fluids. Nevertheless, the macroscale testing sometimes may not be able to characterize the softening mechanism accurately. A reasonable doubt for the large-scale testing is that if the rock-fluid reaction time is too short, the reacted part can be only a very thin exterior layer of the sample due to shale's intrinsic, extremely low permeability, and thus the observed differences in the measured properties could be simply caused or masked by the variability of the tested samples.

As an alternative, non-destructive statistical nanoindentation testing has been recently used for the mechanical characterization of shales at different length scales (Abedi et al., 2016; Liu et al., 2018; Luo et al., 2020; Ulm and Abousleiman, 2006; Zeng et al., 2019). In particular, a big data nanoindentation technique lately developed by Luo et al. (2020) can simultaneously extract the mechanical properties of a shale's individual constituent phases at the nanometer scale as well as bulk rock at micrometer scale based on a single testing technique on a single sample, eliminating the potential errors due to samples' variability and different testing methods' incompatibility. Since nanoindentation only probes a relatively small volume of the material, it has fewer requirements on the size and shape of the tested samples, which provides additional unique advantages for the characterization of shale softening phenomena, such as reducing the costs for sample collection and expediting the rock-fluid interaction process (Akrad et al., 2011; Corapcioglu et al., 2014; Lu et al., 2020; Yang et al., 2018). In this paper, a new technique of screening and optimizing hydraulic fracturing fluid additives enhanced by big data nanoindentation is presented, which is achieved by quantifying and comparing the effects of physicochemical rock-fluid interactions on the mechanical properties of a clay-rich shale and hence understanding the mechanisms of shale softening. Findings show that this new technique can be implemented to screen, design, and optimize the fracturing fluids to be used in different shale formations for enhanced production of hydrocarbons.

2. Materials and experimental methods

2.1. Samples and sample preparation

The studied shale samples were recovered from the Lower Silurian Longmaxi Formation in the Sichuan Basin, China at Latitude 28° 21′ 10″ N and Longitude 103° 57′ 19″ E. The relatively large intact cores were originally used for triaxial testing, and the post-failure fragments from the fractured cores were collected for this study. Five smaller samples with an in-plane (i.e., bedding plane) dimension of ~3 cm and thickness of ~1.5 cm were trimmed from the post-failure fragments. While one sample was tested without fluid pretreatment, the other four were subjected to softening treatment prior to indentation testing. Its mineralogical compositions determined by X-ray diffraction (XRD) were reported in prior work (Luo et al., 2021): (1) a total organic carbon (TOC) of 2 wt%; (2) a porosity of 0.07; (3) a clay (mostly illite) fraction and a nonclay, solid inclusion fraction of 46 and 47 vol%, respectively, the latter of which consists of 55.3 wt% carbonate, 42.6 wt% relatively strong minerals including quartz and feldspar, and 2.1 wt% pyrite.

To prepare highly smooth and flat surfaces for nanoindentation testing, the small cuboidal samples were first mounted onto a cylindrical aluminum puck by a Crystalbond 509 amber resin mounting adhesive with their bedding plane oriented horizontally, followed by polishing the bedding plane in a MetaServ 250 polisher (Buehler Inc., Lake Bluff, IL, USA) in accordance with a well-developed procedure detailed in Luo et al. (2020). According to previous atomic force microscopy measurements, the resulting root-mean-square (RMS) roughness of the polished surfaces can be minimized to ~100 nm.

2.2. Softening treatments

Four different fluids, including deionized (DI) water, 5 wt% aqueous KCl solution, and Surfactant A and Surfactant B solutions (hereafter abbreviated as Surf A and Surf B, respectively), were used as the fracturing fluids to react with the four shale samples, respectively. The former two are common ingredients in nearly all fracturing fluids, whereas the latter two are commercial microemulsions dissolved in 5 wt % KCl base solution and their detailed compositions are confidential. The polished samples were removed off the aluminum puck by melting the resin at 130 °C, and the residual resin attached to the bottom of the samples was completely removed by acetone prior to softening treatments. A two-step softening treatment was developed for each sample: (1) submerging the polished sample in one selected fluid, followed by saturation in vacuum for 24 h; and (2) performing rock-fluid reactions at 2.0 MPa and 121 °C for another 24 h to simulate the deep underground hydrothermal environment where hydraulic fracturing usually takes place. At the end, the reacted samples were air-dried at ambient pressure and temperature, and then the sample was re-glued onto the aluminum puck by the same mounting adhesive.

2.3. Grid indentation testing

The mechanical properties, including Young's modulus E and hardness H, of all five samples were probed by conducting 588 indentation measurements (i.e., a total of 12 matrices, each with 7 × 7 indents) in a G200 nanoindenter (Keysight Technologies, Inc., Santa Rosa, CA, USA) using a customized Berkovich diamond tip with a tip radius of <20 nm and effective depth of 30 μm. A target maximum indentation depth of 8 μm was set for all indents, and the continuous stiffness measurement (CSM) method was used for loading at a constant indentation strain rate ($\dot{h}/h$) of 0.05 s^{-1}, which involved the superimposition of a sinusoidal displacement-controlled signal of 2 nm and 45 Hz in amplitude and frequency, respectively. A relatively large grid spacing of 200 μm between two adjacent indents was selected for such deep indents to avoid negative interference between adjacent indents (Haušild et al., 2016; Sudharshan Phani and Oliver, 2019). Unlike the conventional nanoindentation that can only determine the mechanical properties at the maximum indentation depth, the CSM method enables the continuous measurement of mechanical properties along the penetration depth during the entire loading process by analyzing the dynamic response of the sample-tip-indenter system caused by a small harmonic oscillation superimposed on the primary monotonic loading signal (Li and Bhushan, 2002; Oliver and Pharr, 2004; Saha and Nix, 2002). Further noteworthy is that the untreated sample was previously studied in Luo et al. (2021), and the results are used in this study as the baseline reference.

2.4. Surface characterization

Due to the concern with the potential damage to the surface caused by the energized electron beam, the polished surfaces of the five samples were examined by scanning electron microscopy and energy-dispersive X-ray spectroscopy (SEM-EDS) after nanoindentation testing. A thin layer of conductive carbon coating was applied to the sample surface to eliminate the buildup of electron charging during SEM imaging. Elemental mapping was conducted at two magnifications in a Zeiss EVO50 XVP SEM (Carl Zeiss Microscopy GmbH, Jena, Germany) equipped with a Bruker Quantax EDS (Bruker Nano GmbH, Berlin, Germany) to facilitate phase identification in conjunction with the known mineralogy determined by XRD and to obtain comprehensive chemical data for the near-surface constituents. At the relatively lower magnification of 100×, a larger area of the surface was observed so that the results were less sensitive to the scanned locations, while the high-resolution elemental maps collected at a 400× magnification potentially detected the subtle changes in chemical compositions on or near the sample surface caused by softening treatment. Additionally, high-resolution SEM images were also obtained in a Magellan 400 XHR SEM (FEI Company, Hillsboro, OR, USA) to examine the potential alterations to the surface morphology and microstructure induced by softening treatment.

3. Big data nanoindentation

The big data nanoindentation technique used in this study was originally proposed and validated in Luo et al. (2020), with further improvements in data analytics (Luo et al., 2021). Given the length constraint, this section only briefly introduces this new indentation technique, while further details can be found in the aforementioned two articles. The basic concepts schematically summarized in Fig. 1 will be elaborated in the following sections.

3.1. Continuous property-depth data

Fig. 1a shows the unprocessed Young's modulus-depth (E-h) and hardness-depth (H-h) curves obtained for the untreated shale sample, while the measurements for the other four treated samples can be found in Appendix Fig. S1. For highly heterogeneous materials such as shale, the results directly collected from the grid indentation testing are usually chaotic and highly scattered (e.g., Du et al., 2018; Lu et al., 2020; Luo et al., 2021, 2020; Wu et al., 2020). Similarly, the first observation about the unprocessed E-h and H-h curves is that both the E and H are dependent upon h, and these curves are highly scattered and dispersed over a wide range when the indentation depth is shallow, but gradually converge to a much narrower range with increasing depth. Such depth-dependent mechanical behavior is not uncommon for such heterogeneous materials as shales, because indentation measurements on a multiphase and multiscale composite can be affected not only by the material or volume that is in contact with the indenter tip, but also by the surrounding material that is inside the elasto-plastic zone created by the tip load. Such a phenomenon is called the "indentation surround effect", which describes the dependence of indentation results on the relationships among indentation depth, length scale of different phases or particles, and respective constitutive law of these materials. Further elaboration can be found in the subsequent sections of this paper, and more discussion on this topic can be found in recent literature (Lu et al., 2020; Luo et al., 2021, 2020; Wu et al., 2020).

To clearly show the dependence of indentation measurements on the depth and the so-called "indentation surround effect", Fig. 2 selectively shows the drastically different responses of three indents made on three spots with different stiffnesses, at least on or near the surface. Their E-h and H-h curves are separated apart at shallow depths (e.g., $h < 2$ μm), but merge to a nearly constant at large depths (e.g., $h > 4$ μm). Such a phenomenon clearly manifests the indentation location-dependent measurements, and manual separation of the >500 E-h and >500 H-h curves is challenging or even infeasible (Fig. 1a). Moreover, a clear boundary between the curves of stiffer phases and those from the softer counterparts may not exist. As such, the corresponding big data analytics was developed to process these highly scattered, chaotic measurements, and implementing proper data analytics becomes essential for correctly processing the massive indentation measurements, in order to extract meaningful and definitive properties of the tested samples.

3.2. Big data analytics

Since shale is a multiphase and multiscale composite, constant mechanical properties measured over different indentation depths are not usually expected (e.g., Manjunath and Jha, 2019). The big data nanoindentation technique also involves big data analytics, a viable four-step framework consisting of data segmentation and extraction, statistical deconvolution, data re-assembly, and surround effect modeling, which leads to the extraction of the mechanical properties of individual phases at the nano/micro scale as well as the progressive transition to the counterparts of large-scale bulk rock.

To achieve cross-scale characterization, the unprocessed continuous E-h and H-h curves were first segmented at an interval of 0.05 and 0.1 μm for the h of ≤1.0 and >1.0 μm, respectively (Fig. 1a), leading to the generation of approximately 50 two-dimensional (2D, i.e., each data entry contained a pair of E and H values extracted from the same depth) subdatasets for each tested sample (Fig. 1b). Next, each subdataset was statistically analyzed by Gaussian mixture modeling (GMM), a widely used method for processing grid indentation results (Akono and Kabir, 2016; Krakowiak et al., 2015; Wilson et al., 2018). Provided that the tested sample consists of k mechanically distinct phases whose Young's modulus and hardness obey a bivariate Gaussian distribution g, the theoretical probability density function (PDF) of the mechanical properties of this k-phase composite is as follows:

$$f_k(\boldsymbol{x};\boldsymbol{\Psi}) = \sum_{i=1}^{k} \pi_i \, g(\boldsymbol{x};\boldsymbol{\mu}_i,\boldsymbol{\Sigma}_i) \tag{1}$$

where $\boldsymbol{x}$ is the 2D subdataset (H, E) extracted a given segmentation depth; π_1, …, π_k are non-negative mixing proportions whose sum is

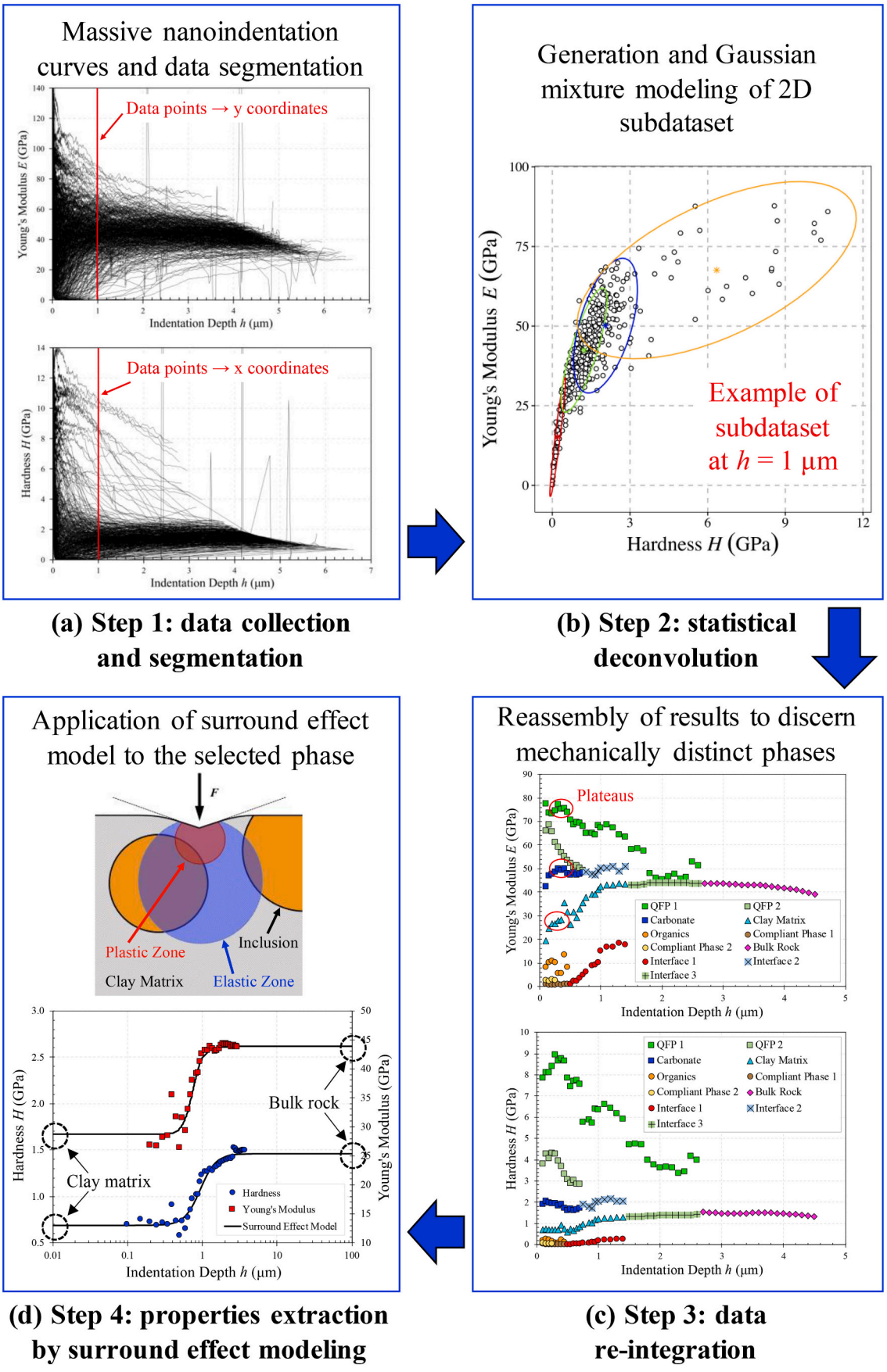

Fig. 1. Schematic flow diagram illustrating the fundamental principles of big data nanoindentation technique and surround effect modeling; note that the results for untreated shale are adopted from Luo et al. (2021).

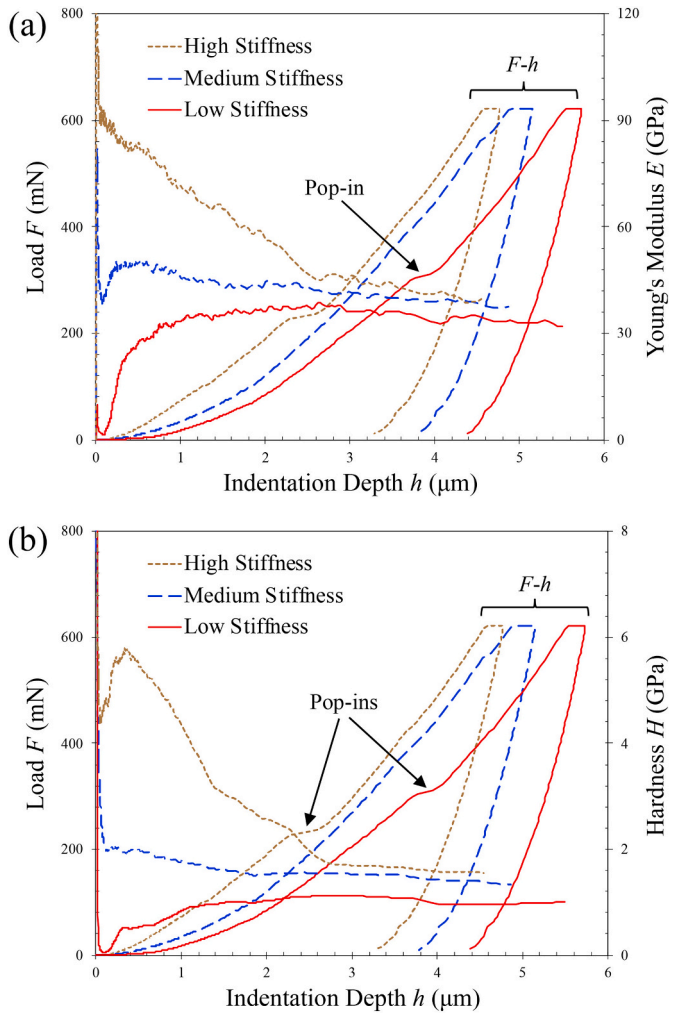

Fig. 2. Indentation measurements for three phases with distinct stiffness: (a) load-depth and Young's modulus-depth curves; and (b) load-depth and hardness-depth curves. Depth-dependent mechanical properties are observed for all exhibited phases.

equal to 1.0; $\boldsymbol{\mu}_i$ and $\boldsymbol{\Sigma}_i$ denote a 2 × 1 mean vector and a 2 × 2 variance-covariance matrix of the i^{th} phase's Gaussian distribution, respectively; $\boldsymbol{\Psi}$ is the set of parameters $\boldsymbol{\Psi} = (\pi_1, \ldots, \pi_{k-1}, \boldsymbol{\mu}_1^T, \ldots, \boldsymbol{\mu}_k^T, \ldots, \text{vech}(\boldsymbol{\Sigma}_1)^T, \ldots, \text{vech}(\boldsymbol{\Sigma}_k)^T)^T$ where vech($\boldsymbol{A}$) means that only the lower triangular part of a symmetric matrix $\boldsymbol{A}$ is vectorized (i.e., half-vectorization). The unknown parameters $\boldsymbol{\Psi}$ were fitted by the maximum likelihood estimation (MLE) via expectation-maximization (EM) algorithm using an R programming package called "EMMIX" (McLachlan et al., 1999). Generally, the true number of mechanically distinct phases, *k*, in a shale is unknown, but its upper bound can be determined according to the mineralogical analysis such as quantitative XRD.

In this study, the experimental data were fitted by eight different Gaussian mixture models $f_k(\boldsymbol{x}; \boldsymbol{\Psi})$ with *k* ranging from 1 (i.e., lower bound) to 8 (i.e., upper bound), and the parameters $\boldsymbol{\Psi}$ of each model were randomly initialized 1000 times because the maximum likelihood estimator of $\boldsymbol{\Psi}$ might not be the same each time if the parameters were initialized differently. Therefore, a total of 8000 fittings were conducted for each subdataset, resulting in ~400,000 fittings for each sample.

Multiple unique solutions were generally obtained for models with $k > 1$ after the above data fitting process. A significant advantage of the GMM is that some likelihood-based methods such as Akaike's information criterion (AIC) (Akaike, 1998), Bayesian information criterion (BIC) (Schwarz, 1978), and integrated completed likelihood (ICL) (Biernacki et al., 2000) are available for assisting the selection of the best model from the statistical perspective (Kim and Seo, 2014; Seo and Kim, 2012). Therefore, the next step of big data analytics was to find the best one out of all available fitting solutions for each segmented subdataset. For instance, Fig. 1b is a plot of the statistically analyzed result for the subdataset extracted at the indentation depth of 1.0 μm. Four phases with distinct mechanical properties were objectively identified by the GMM, together with the BIC. Consequently, the new *E-h* and *H-h* curves can be generated for each constituent phase of the shale by reassembling the means of *E* and *H* of each Gaussian distribution along the indentation depth, which actually reversed the first step of data segmentation and re-established the relationship between each specific mechanical property and indentation depth (Fig. 1c). More details on the GMM-based big data analytics can be found in Luo et al. (2021), especially the solution selection criteria and the methods of data re-assembly.

3.3. Cross-scale characterization of mechanical properties

Fig. 1c presents the cross-scale mechanical properties of the shale derived from the originally chaotic and highly scattered *E-h* and *H-h* curves (Fig. 1a) via the big data analytics. When the indentation depth is small (e.g., $h < 0.5$ μm), the indentation load only interacts with a small volume just beneath the sample surface, and thus the mechanical properties of individual constituent minerals are usually probed at this scale (Constantinides et al., 2003; Ulm et al., 2007). Seven phases with distinct mechanical properties are identified for the untreated sample at the beginning of the loading process (i.e., at small indentation depths). Based on the knowledge of the mineralogical composition of the sample and the published literature in the stiffness of different minerals (Mavko et al., 2009), the above mechanically-distinct phases can be classified into four major groups, including QFP (i.e., quartz, feldspar, and pyrite), carbonate, clay matrix, and organic matter, in the descending order of mechanical properties (i.e., Young's modulus, hardness). Since the QFP includes three different minerals with similar stiffness but very different particle sizes, it is further divided into two sub-groups labeled as QFP 1 and QFP 2. The clay matrix, but not the clay mineral phase, is specifically defined as a homogenized composite consisting of: (1) preferentially oriented, highly compacted, platy particles (<2 μm in size) of different clay minerals (e.g., illite, muscovite, and chlorite); (2) relatively small (i.e., <1 μm in size) carbonate particles as the cementing agent for the clay matrix; and (3) nanopores among the clay particles. Moreover, additional phases labeled as the "compliant phases" are not any real constituents of the shale but actually virtual phases that represent the measurements from a small group of indents made on pores or asperities randomly distributed on the polished sample surface (de Souza et al., 2006; Li et al., 2009). As such, they only occur at small indentation depths and usually have a very low Young's modulus and hardness.

Unsurprisingly, the depth dependence of the Young's modulus and hardness exhibited in the unprocessed data (Fig. 1a) is also reflected in the results processed by the GMM-integrated big data analytics (Fig. 1c). Essentially, the mechanical properties sensed by an indenter is the averaged mechanical response of all different phases inside the elasto-plastic zone (i.e., the indenter-sample interaction volume) induced by indentation load (Fig. 1d). With continuously increasing indentation depth, the size of the elasto-plastic zone beneath the indenter tip keeps on expanding and eventually exceeds the characteristic length scale (i.e., the size) of the initially indented mineral, leading to the dependence of the mechanical properties on indentation depth (Luo et al., 2020). For example, the mechanical properties of the stiffer phases, including QFPs and carbonate, start to decrease when the depth increases beyond the initial 0.3–0.5 μm. Compared with the QFP 1, the mechanical properties of QFP 2 start to decrease even at a smaller depth and a higher rate due to its relatively smaller particle sizes, and then the lack of contrast in their mechanical properties leads to the combination of QFP 2 and carbonate phases by statistical analysis at the depth of 0.75 μm, accompanied by the generation of a new intermediate phase, Interface 2,

demonstrating the locally averaged multiphase responses. Such progressive homogenization of mechanical properties with increasing depth is not uncommon, especially for highly heterogeneous materials such as shale, sandstone, and concrete.

Moreover, the mechanical properties of other three softer constituent phases also show a similar depth-dependent behavior but in the opposite trend (i.e., increasing with depth). Those new phases (i.e., Interfaces 1 to 3), created by joining two different neighboring phases, further converge at much larger indentation depth, leading to the formation of a unified single phase with approximately constant or depth-independent mechanical properties. In fact, the unified single phase reflects the bulk rock's properties, because the elasto-plastic zone at this later stage of very large indentation depth is sufficiently large to encompass a representative volume of bulk sample within which all different individual phases are included, and the measurement reflects the combined responses of all phases or the bulk composite.

Similar to the untreated shale, the depth-dependent mechanical properties and the multi-staged converging behavior are also observed in all four hydrothermally treated samples (Appendix Figs. S2 and S3). The only difference in the phase classification for these treated samples is that the organic phase is merged into the compliant phase, since the organic matter in shales generally becomes softer at the elevated temperatures of 150 °C or higher (Emmanuel et al., 2016). According to previous studies, the overall reduction in Young's modulus or hardness caused by softening can be as high as 70%, depending on the reaction environment (Akrad et al., 2011; Corapcioglu et al., 2014; Das and Achalpurkar, 2013). However, the depth-dependent behavior uncovered in Figs. S2 and S3, which is critical for the quantitative evaluation of shale softening, has not been addressed previously. This missing information in the literature may lead to a possible consequence that the properties of different phases before and after treatment are compared, since these properties are only obtained from a fixed indentation depth but without the depth-dependent transition. A simple example related to this concern is that, if a fixed, constant indentation depth of 0.5 μm is used to determine and then compare the Young's modulus of clay matrix in different samples (Fig. 3), the effects of different fracturing fluids on shale softening are likely to be overestimated, because the baseline Young's modulus of the clay matrix for the untreated sample is already higher than the true value normally determined from the initial horizontal section or plateau (i.e., a relatively constant value at smaller depths) of the curves (e.g., the example plateaus are shown by red circles in Fig. 1c).

3.4. Indentation surround effect modeling

Since the clay matrix is the major load-bearing medium of fine-grained shales, which dominates the mechanical properties of not only the bulk rock, but also other minerals embedded therein, understanding how the mechanical properties of clay matrix change before and after hydrothermal treatments is the most crucial for assessing the degree of softening and hence functional performance of chemical additives. Therefore, the mechanical properties of the clay matrix (i.e., the load-bearing phase) and its subsequently derived phases (e.g., from Interface 2 to bulk rock) are fitted by the indentation surround effect model (Luo et al., 2020; Wei et al., 2009) described below:

$$M = M_b + \frac{M_c - M_b}{1 + (h/a)^Y} \quad (2)$$

where M is a kind of mechanical property (i.e., either Young's modulus or hardness) determined by nanoindentation at a specific indentation depth of h (i.e., the individual data points in the plots of Fig. 1c); M_c and M_b are the pertinent mechanical properties of the clay matrix and bulk rock, respectively. Together with the other two parameters a and Y, they are determined through the curve fitting.

Fig. 1d shows the semi-logarithmic plots of both Young's modulus and hardness versus indentation depth for the clay matrix phases of the untreated sample, including the discretized data points derived from the big data analytics (Fig. 1c) and the pertinent curves fitted by the indentation surround effect model (Eq. 2). It is observed that the model enables a smooth and progressive transition from individual phase-dominated response to bulk rock-dominated one as the indentation depth increases, a phenomenon called "cross-scale characterization". More importantly, some detrimental factors can be overcome by implementing such model fitting. For example, the mechanical

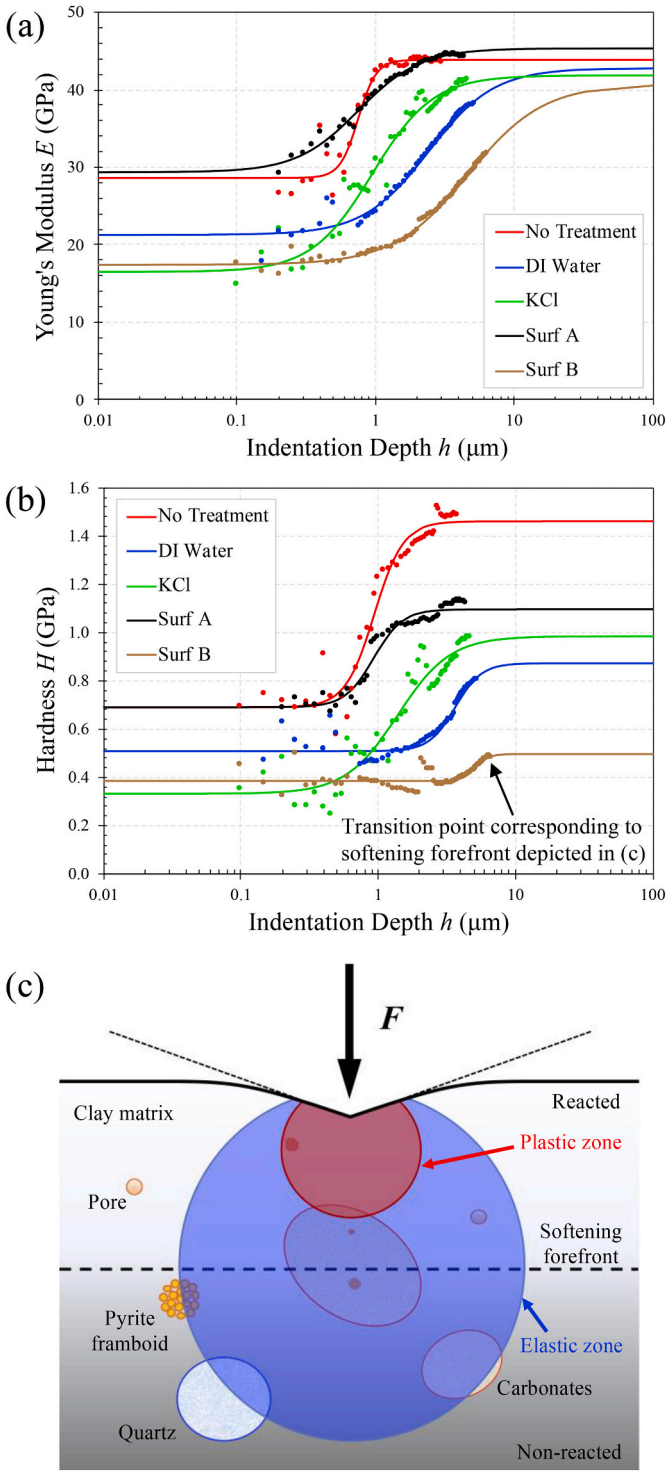

Fig. 3. The nonlinear regression of the depth-dependent Young's modulus (a) and hardness (b) of clay matrix using the surround effect model (c).

properties of the clay matrix are more accurately determined from the fitted curves at very small depths (i.e., 0.1 μm) even if the effect of sample's surface roughness is unavoidable in the unprocessed experimental measurements; conversely, the bulk rock's properties are determined at the very large depth of >10 μm, eliminating the limitation imposed by the maximum load capacity of the instrument (i.e., a 10 μm indentation depth may not be achievable if the sample is very strong and the indenter has a limited maximum load).

To directly compare the different mechanical properties of the shale before and after hydrothermal treatments, the model-fitted results of different samples are plotted in Fig. 3a and b. The mechanical properties of clay matrix and bulk rock are also summarized in Table 1, along with those of QFP and carbonate phases determined from Figs. 1c, S2, and S3. It is worth pointing out that the Young's modulus and hardness of these two stiffer phases are calculated regularly by averaging the plateaus occurring at the small depths. If there is no such a pronounced plateau, then the highest value is selected instead.

4. Screening of chemical additives

4.1. Indentation measurements

In all the *E*-*h* and *H*-*h* curves presented in Fig. 1a and Fig. S1, the maximum indentation depth, even for the indents made on the same sample surface, varies considerably and is usually less than the preset target of 8 μm. Fig. 4 shows how the normalized number of indentation measurements (i.e., the ratio of the number of available data points in each subdataset to the total number of indents) changes when the depth increases, where all curves have a relatively flat section at the beginning. The turning point on each curve in Fig. 4 marks the depth at which the number of effective measurements starts to decline significantly. In fact, the reduced size of subdatasets after the turning point does have a slight influence on the results of statistical analysis, which is reflected by the slight decrease in the mechanical properties of the bulk rock at larger depths (e.g., Fig. 1c).

The reason why most indent measurements did not reach the preset maximum displacement is that these indents were made on the relatively stronger mineral phases with a relatively high stiffness and hardness (e.g., quartz, feldspar, and carbonate), and hence the equipment's maximum load capacity (i.e., 600 mN) was reached prior to the target maximum depth, which stopped the measurement prematurely. As such, these turning points are qualitative indicators of the overall stiffness or mechanical properties of the samples. In other words, the stiffer the sample, the shallower the depth of the turning point. In fact, the sample without any treatment (i.e., the baseline reference) has the shallowest turning point, while the depths of turning points for all other samples increase to various extents. Therefore, an intuition here is that different hydrothermal treatments more or less alter the overall stiffness of the same shale.

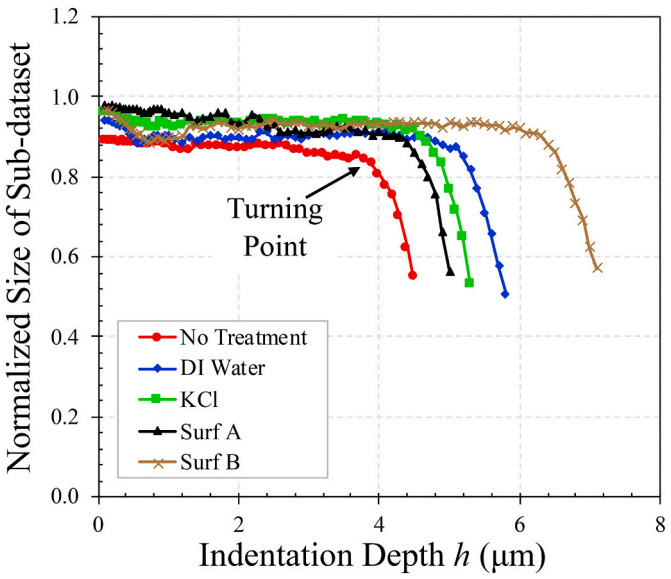

Fig. 4. The normalized size of subdataset versus the indentation depth.

4.2. Mechanical properties of individual phases

Based on Fig. 3a–b and Table 1, the mechanical properties of the clay matrix (i.e., extracted at $h = 0.01$ μm) significantly decrease upon the hydrothermal treatment by different fracturing fluids, except the Surf A. Specifically, the DI water treatment reduces the *E* and *H* of the clay matrix by 25.8% and 28.6%, respectively, although the shale sample contains no swelling clay minerals but illite and chlorite. Also, the treatment by the KCl solution is the worst one, which causes the *E* and *H* of the clay matrix to decrease almost twice as much as those in the DI water-treated sample. Fig. 5a and b compare the SEM images of two residual indents selected from the samples without treatment and with treatment by the KCl solution, respectively. Both indents were made on the clay matrix, but different surface features are observed. For the untreated sample, neither radial cracks nor layer delamination is observed, indicating a conformal deformation of the clay matrix to the indenter tip. However, after the KCl solution treatment (Fig. 5b), the regions near the top and lower left of the residual indent heave up significantly through layer delamination and basal plane rupture (Zhang et al., 2013), suggesting that the structural integrity of clay matrix is significantly undermined by the shale-fluid interactions. Moreover, Fig. 3 shows that the two used surfactants have very different effects on the change in mechanical properties of the clay matrix. For example, both the *E* and *H* of the clay matrix in the Surf A-treated sample remain nearly the same as those of the untreated counterpart. However, very little improvement in the mechanical properties is found for the Surf B

Table 1
Summary of the mechanical properties of individual phases and bulk rock obtained from the big data-based nanoindentation.

Fluid Type	*E* in GPa (Reduction in %)[1]							*H* in GPa (Reduction in %)[2]			
	Clay Matrix	Carbonate	QFP	Bulk Rock	VRH	MT	SC	Clay Matrix	Carbonate	QFP	Bulk Rock
None (Baseline)	28.7 (0)	48.5 (0)	72.0 (0)	43.8 (0)	44.0 (0)	43.4 (0)	43.6 (0)	0.7 (0)	1.9 (0)	6.5 (0)	1.5 (0)
DI Water	21.3 (25.8)	32.7 (32.6)	57.0 (20.8)	42.9 (2.1)	28.4 (35.5)	27.8 (35.9)	28.0 (35.8)	0.5 (28.6)	1.4 (26.3)	4.5 (30.8)	0.9 (40.0)
5% KCl	16.5 (42.5)	29.8 (38.6)	51.2 (28.9)	41.9 (4.3)	29.7 (32.5)	28.8 (33.6)	29.2 (33.0)	0.3 (57.1)	0.9 (52.7)	4.6 (29.2)	1.0 (33.3)
1 gpt Surf A in 5% KCl	29.3 (−2.1)	46.0 (5.2)	62.8 (12.8)	45.3 (−3.4)	37.1 (15.7)	36.8 (15.2)	36.9 (15.4)	0.7 (0)	1.8 (5.3)	6.9 (−6.2)	1.1 (26.7)
1 gpt Surf B in 5% KCl	17.4 (39.4)	24.4 (49.7)	46.6 (35.3)	40.7 (7.1)	26.6 (39.5)	25.9 (40.3)	26.1 (40.1)	0.4 (42.9)	0.8 (57.9)	3.8 (41.5)	0.5 (66.7)

[1,2] The percentage reduction in Young's modulus *E* and hardness *H* is calculated based on the baseline results, and negative values mean increase in mechanical properties after treatment.

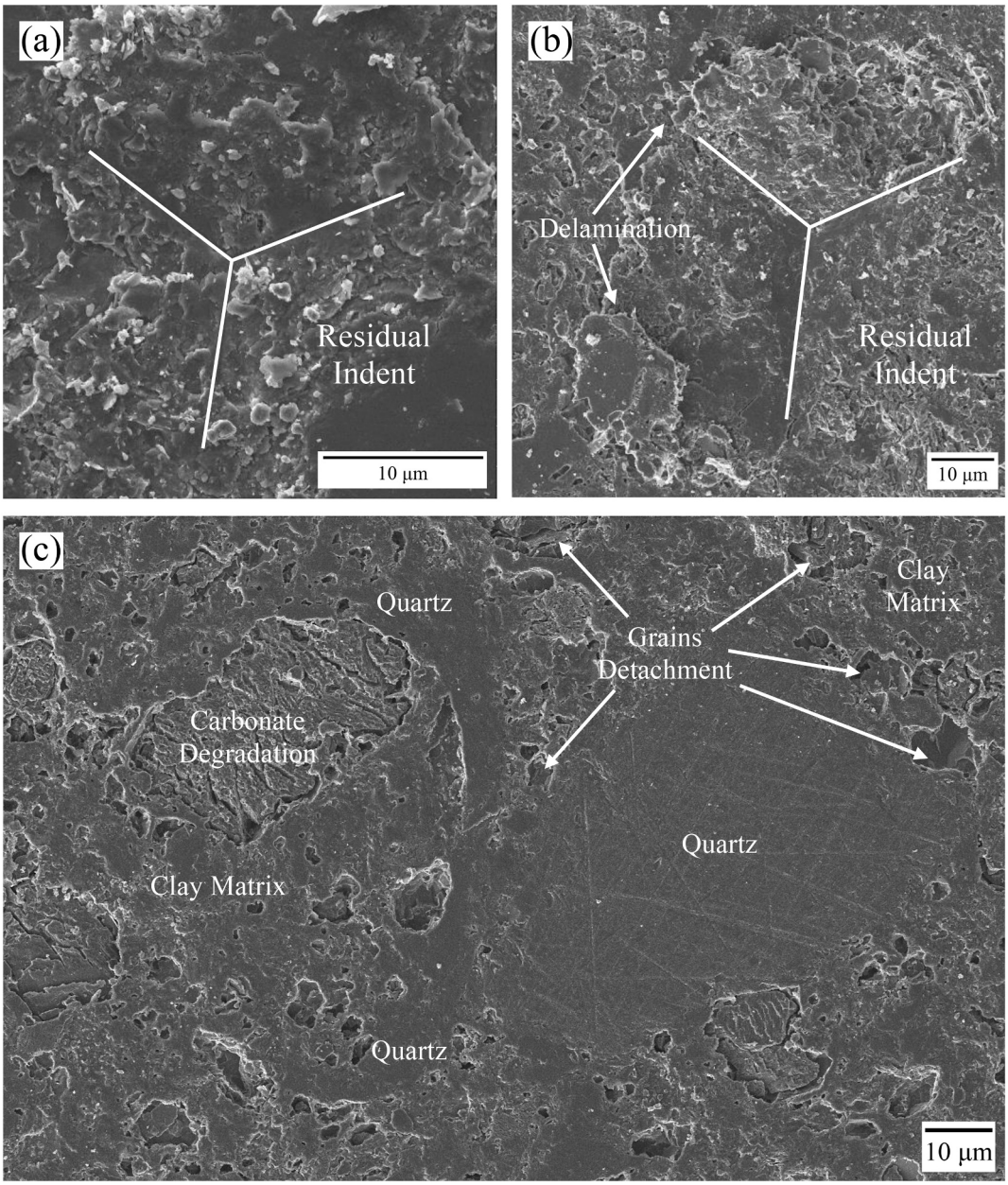

Fig. 5. The SEM micrographs showing: (a) a residual indent obtained from the untreated sample; (b) a residual indent obtained from the KCl-treated sample; and (c) a region from the DI water-treated sample containing relatively large quartz (not reactive to treatment) and carbonate particles (reactive to treatment).

when compared with those obtained from the 5% KCl solution (i.e., the base solution used for the two surfactants).

In Table 1, it is interesting to note that the mechanical properties of all relatively stiff inclusions, including the QFP and carbonate phases, also decrease after the hydrothermal treatments, despite that quartz and feldspar are often considered to be very stable and resistant to different chemical additives (Fig. 5c). In fact, the indentation surround effect suggests that the measurements from the relatively small solid inclusions (e.g., quartz) are not only controlled by the stiffness and strength of the indented material itself, but also affected by the particle size and the stiffness mismatch between the solid inclusions and soft surrounding clay matrix. Here, the softening of the clay matrix actually exacerbates such a mechanical mismatch, which causes the measured mechanical properties of inclusions to be affected by the softened clay matrix literally at the beginning of the indentation loading (Luo et al., 2021). For the carbonate, the chemical degradation shown in Fig. 5c is another reason responsible for the reduction in their mechanical properties, such as the dissolution and defects introduced by the rock-fluid interactions, which can substantially lower their mechanical properties, as demonstrated by the indentation measurements.

4.3. Mechanical properties of the bulk rock

As discussed previously, the cross-scale characterization and indentation surround effect model enable the determination of mechanical properties of bulk rock at the micrometer scale. The second transition point at the larger depths on each continuous curve in Fig. 3a and b is actually the critical depth beyond which the bulk rock's properties can be obtained. Generally, based on the 10% rule of thumb (Fischer-Cripps, 2011), the size of the indentation-induced plastic zone (responsible for the hardness measurement) is approximately ten times greater than the indentation depth, thus the maximum achievable size of the plastic zone for each sample can be estimated by multiplying the critical depth by a factor of ten. Since the bulk rock's hardnesses of the four treated samples

are significantly lower than that of the untreated counterpart (Table 1), the critical depth (i.e., transition point indicated in Fig. 3b) is a conservative indicator of the size of the softening zone. For instance, the DI water-treated sample has a critical depth of ~7 μm, indicating that the depth of softening reaction has advanced at least 70 μm underneath the sample surface during a two-day treatment (i.e., at an advancing rate of >35 μm/day). For comparison, Lu et al. (2020) reported a large volume-based nanoindentation technique for assessing the softening behavior of shales in the DI water, and a softening advancement rate of 10.5 μm/day was quantitatively determined. The faster softening advancement rate (i.e., 35.0 versus 10.5 μm/day) obtained in this study is probably caused by the higher pressure (i.e., 2 MPa versus atmospheric pressure) used for the rock-fluid interaction experiments.

In contrast to hardness, there is no substantial distinction in the Young's modulus of the bulk rock when subjected to different treatments (i.e., the difference is <7.1% according to Table 1). This is due to the fact that the elastic zone induced by indentation loading is much greater than the plastic zone, and hence a considerable volume of the unreacted or non-softened sample may be encompassed in the elastic zone that contributes to the Young's modulus measurements when the bulk rock is probed at the larger depths (e.g., the blue circle in Fig. 3c). It also proves that a very short-term (e.g., a few days) treatment may only alter the properties of a thin surface layer (e.g., at the micrometer scale) of the shale that usually has a very low permeability. Therefore, a misleading conclusion that shales are immune to the investigated fracturing fluids may be drawn in such circumstances if data interpretation is solely based on the Young's modulus of bulk rock shown in Fig. 3a, but not other properties or the individual phases, particularly carbonate and the clay matrix. For the same reason above, application of the traditional large-scale mechanical measurements such as triaxial testing to the study of shale softening should also be cautious, and the interpretation of results may require careful scrutiny.

Since both the Young's modulus and volumetric fraction of each constituent phase, including mainly the clay matrix, carbonate, and QFP, are determined by the big data nanoindentation method, it is logical to estimate the bulk rock's Young's modulus using some published, widely-accepted homogenization and upscaling models (Kong et al., 2019; Luo et al., 2021; Sebastiani et al., 2016). In this study, three different models applicable to the matrix-inclusion type of mixtures or composites are used, including Voigt-Reuss-Hill (VRH) average (Hill, 1952; Reuss, 1929; Voigt, 1889), Mori-Tanaka (MT) method (Benveniste, 1987; Mori and Tanaka, 1973), and self-consistent (SC) approach (Berryman, 1995; Berryman, 1980). In general, the three methods yield very similar results (Table 1). For the untreated sample used as a baseline reference, all three models are capable of providing very close predictions (i.e., within a less than 2% difference) of the Young's modulus of bulk rock, indicating that the model-based mechanical homogenization and upscaling perform well in this shale sample. Furthermore, the aforementioned interference from the non-reacted part of the sample on the experimental measurements (Fig. 3c) can be avoided as expected in those hydrothermally treated samples, and a clear reduction trend that is consistent with the change in the bulk rock's hardness is also obtained after upscaling.

In addition to the Young's modulus and hardness, fracture toughness has also been recognized as an effective proxy for assessing fracture-related problems (e.g., crack initiation and growth) in shales (Li et al., 2019; Manjunath and Jha, 2019). Recent advances in an energy-based method allows the direct estimation of fracture toughness from indentation load-displacement measurements (Liu et al., 2016; Zeng et al., 2019), and the results show that the fracture toughness generally increases linearly with Young's modulus (Mahabadi et al., 2014). The same positive correlation was also found between the fracture toughness and the combined fraction of TOC, clay, and calcite (Akono and Kabir, 2019). This in fact is consistent with the findings that a higher fraction of calcite promotes interparticle cementation and hence strengthens the mechanical properties of shale (Luo et al., 2021). Therefore, a reduction in Young's modulus induced by the rock-fluid interactions is usually accompanied by a decrease in fracture toughness and the mechanical weakening of the rock's load-bearing skeleton or matrix (Fuchs et al., 2019). The weakened mechanical properties, in conjunction with the presence of cavities and secondary pores, will undoubtedly impair the stability of the wellbore (Josh et al., 2012).

In summary, the big data nanoindentation technique is accurate and sensitive enough to detect subtle changes in the mechanical properties of shale at different scales, and thus can be used as a practical technical toolkit to screen and identify which chemicals can be selected as an additive in a particular fracturing fluid to retard shale softening, or to evaluate the degree of softening if an additive has to be used for other purposes. In this study, an interesting finding is that, besides its positive effect on suppressing swelling of clay minerals, KCl may also negatively impact the mechanical performance of the clay matrix through other interactions and reactions (e.g., dissolution of carbonate), which is to be discussed in the next section. Therefore, as a general rule, KCl is not required as an additive if no water-sensitive clays exist in the reservoir formations (Houston et al., 2009). In fact, even when the target rock formation of hydraulic fracturing contains extensive amounts of swelling clays, additional examination should be performed to assess the positive and negative impacts of different clay-control additives so that a compromised and optimized balance can be achieved between controlling the swelling of clays and reducing the softening of the bulk rock.

5. Softening mechanisms

5.1. Morphological and chemical characteristics

The SEM micrographs obtained from one untreated and four hydrothermally treated samples are shown in Fig. 6. Although these images were not obtained from the same location or surface but the same piece of rock, it can still be clearly observed that the surface morphology of the samples was substantially altered by the softening treatments. The untreated surface is flat, and the clay matrix is densely compacted without visible pores or cavities at this magnification (Fig. 6a), whereas for those samples having been treated with the DI water, KCl solution, and Surf B (Fig. 6b, c, and e), cavities and grooves with sizes ranging from as small as <1 μm to as large as >10 μm are found throughout the surface, indicating that these hydrothermal treatments have generally induced a rougher surface. In addition, as discussed previously (Fig. 5c), the morphologies of different constituent phases after polishing may be completely different even for the same sample. Quartz, as the most common and stable mineral in shales, preserves a flat and smooth surface achieved from the surface polishing, but severe surface degradation is observed for the carbonate particles whose integrity is compromised, indicated by a much rougher, hilly-gully morphological characteristics. In contrast to these solid inclusions, although the small cavities and secondary pores with varying sizes are also found at random locations within the clay matrix (Fig. 5c), it still keeps the overall integrity of the microstructure and maintains a relatively smooth surface.

The mineral distributions on the untreated and treated sample surfaces are further revealed by the EDS elemental mappings (Fig. 7). The examined area shown in Fig. 7 were randomly selected and not spatially correlated to those previously shown in Fig. 6. It can be reassured from the untreated sample (Fig. 7a) that the clay matrix with a light-green color is the load-bearing medium for the tested shale, and the relatively larger silt-sized or sand-sized particles (e.g., quartz in yellowish orange and carbonate in blue colors) are randomly embedded as isolated solid inclusions in a relatively homogenized clay matrix. Besides the large-sized solid inclusions, some carbonate particles also dispersedly occur in the form of tiny particles as the cementation agent contributing to the interparticle bonding within the clay matrix. However, the elemental maps of the treated sample surfaces show that the fraction of carbonate particles significantly decreases after treatments with the DI water, KCl solution, and Surf B, particularly for those smaller cementing

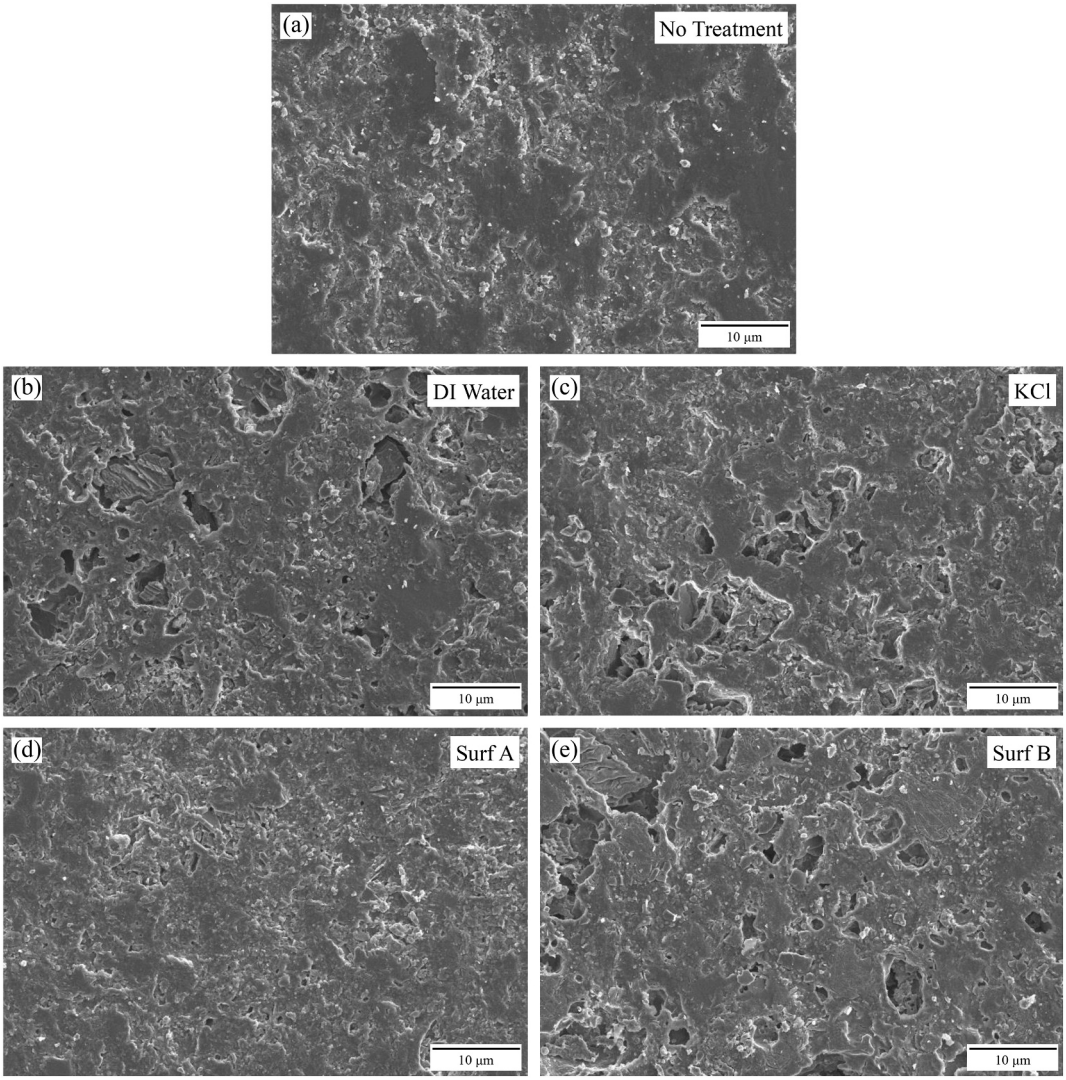

Fig. 6. The SEM micrographs of both untreated and treated shale samples displaying different surface morphologies (at 2500× magnification).

carbonate particles (Fig. 7b, c, and e).

To further verify such observations, elemental maps covering a much larger surface area of 3.2 mm^2 (3.6 × 0.9 mm) were collected from a randomly chosen location of each sample surface (Fig. 8), demonstrating that the aforementioned alterations to surface morphologies induced by hydrothermal treatments are representative of the entire sample surface but not of the specific locations. For the untreated and Surf A-treated samples (Fig. 8a and d), the background color of the clay matrix over a large field of view (FOV) is a mixture of green and blue spots, indicating that the clay matrix contains a relatively high fraction of carbonate cementation. On the contrary, the clay matrix in the other three samples (Fig. 8b, c, and e) remains the same pure green color as that perceived in Fig. 7, meaning that the carbonate cementation within the clay matrix is rarely detected by the EDS after these hydrothermal treatments.

The above EDS results indicate that the surface defects (i.e., cavities and secondary pores) are mainly caused by the disappearance or dissolution of carbonate, such as calcite ($CaCO_3$) and dolomite [$CaMg(CO_3)_2$], after these hydrothermal treatments. Nevertheless, together with the EDS chemical analysis presented above, the change in the microstructure is not evident in the sample treated by Surf A, which only exhibits a limited number of small pores (mostly <2 μm) in the clay matrix (Fig. 6d). Therefore, by comparing with other treatments, Surf A shows the ability to prevent carbonate dissolution to the greatest extent and has the best performance of protecting the microstructural integrity of the shale even under elevated pressure and temperature.

5.2. Pyrite oxidation and dissolution of carbonate

One important chemical reaction occurring during the rock-fluid interactions is pyrite oxidation, which may induce other subsequent reactions. Pyrite (FeS_2), ubiquitous in shales and other sedimentary rocks, is identified in this shale and tends to oxidize upon exposure to air or oxygen and water, leading to the production of ferric iron hydroxide [$Fe(OH)_3$] and sulfuric acid (H_2SO_4) (Golez and Kyuma, 1997). Although pyrite oxidation may involve several chemical reactions divided into different steps, the widely-accepted, overall summary reaction is as follows:

$$4\,FeS_2 + 15\,O_2 + 14\,H_2O \rightarrow 4\,Fe(OH)_3 + 8\,H_2SO_4 \quad (3)$$

The reaction product, sulfuric acid (H_2SO_4), in turn causes dissolution of carbonate cementations (Rimstidt and Vaughan, 2003). It was also pointed out that high concentrations of calcium, magnesium, and sulfate were detected in the supernatant after steeping a crushed shale sample in water for three weeks (Ali and Hascakir, 2017). In this study, an indirect evidence of pyrite oxidation is that the Surf A-treated sample has a bulk rock hardness that is 26.7% lower than that of the untreated sample. Because quartz is relatively inactive and meanwhile the clay matrix and carbonate are well protected by the Surf A added to the KCl

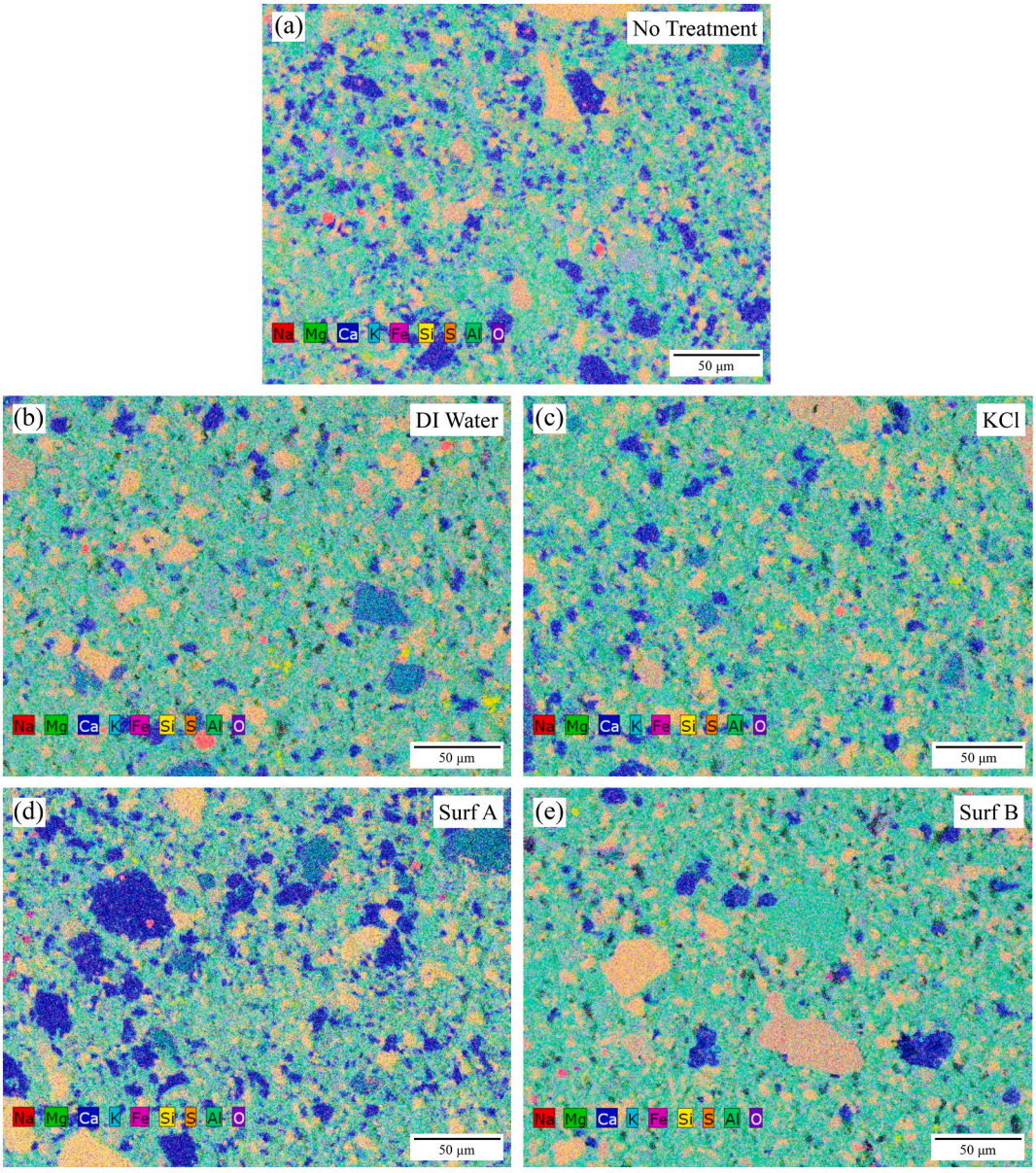

Fig. 7. Elemental mappings collected from the five shale surfaces at the 400× magnification: clay matrix (light green), carbonate (blue), quartz (yellowish orange), and pyrite (orange). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

solution, the decrease in the bulk rock's hardness can then be attributed to pyrite oxidation and subsequent dissolution of carbonate cementation.

Another possible reason accounting for the dissolution of carbonate is the potential physicochemical reactions induced by the KCl solution, because the solubility of calcium carbonate in 5% KCl solution at 25 °C is six times greater than that in water (i.e., 0.078 g/L vs. 0.013 g/L) (Cameron and Robinson, 1907). Besides, as a consequence of carbonate dissolution, the particles with a shallow embedment may lose their physical constraints at the boundaries, allowing the interparticle repulsive forces to push them out of the surface. As such, particle detachment occurring during the rock-fluid interactions is also responsible for the formation of micrometer-sized cavities (Levenson and Emmanuel, 2017; Røyne et al., 2015), which may then promote the formation of certain open microchannels that favor water infiltration into the shale (Deng et al., 2017; Gupta et al., 2017).

5.3. Clay swelling

Although the dissolution of carbonate and other cementation agents, as previously discussed, is a major factor contributing to shale softening, swelling caused by the complex physico-chemical interactions between the clay minerals in shale and fracturing fluids is another important process that usually results in the instability of the rocks. While illite and chlorite identified in the studied shale are known as non-expandable clay minerals that do not exhibit interlayer swelling, osmotic swelling, which is based on the repulsion between the electrical double layers (EDL), can still take place because of the large difference in ion concentrations between the pore fluid close to the clay surface and the bulk pore fluid (Madsen and Müller-Vonmoos, 1989; Sebastián et al., 2008). As a result, the water taken up by the clay particles may act as lubricant and hence reduces the internal frictional resistance of mineral particles (Lu et al., 2020). Moreover, the overlapping double layers can also produce the electrostatic repulsion between adjacent clay particles, favoring the buildup of pressure to the pores and cracks in shale (Du et al.,

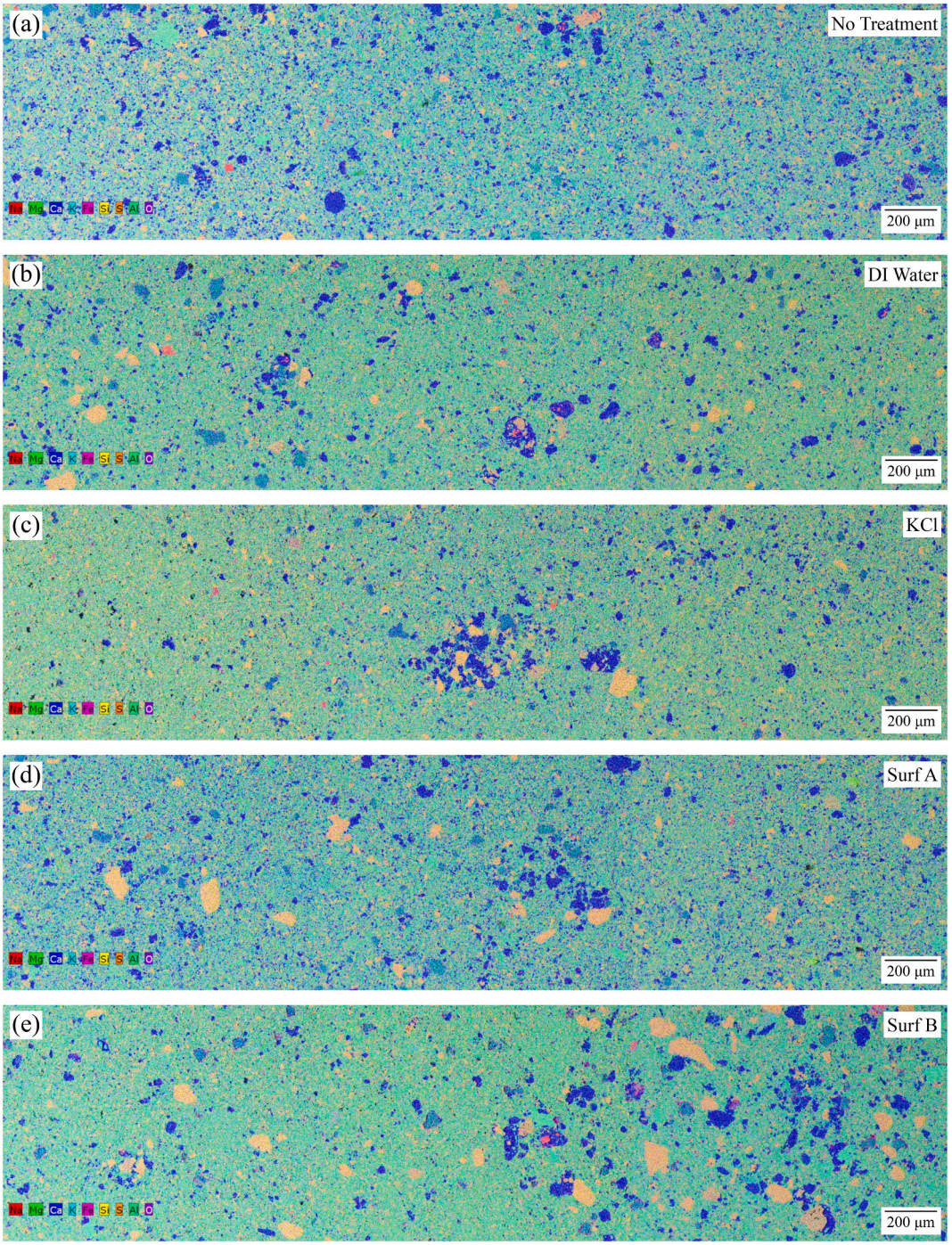

Fig. 8. The surface elemental mappings of the five tested shale samples acquired at a lower magnification of 100× to cover a larger surface area (length 3.6 mm × width 0.9 mm).

2018).

6. Conclusions

In this paper, a recently developed big data nanoindentation technique is applied to characterize the mechanical properties of a shale when subjected to hydrothermal treatments by four different hydraulic fracturing fluids containing various chemical additives. Combined with surface imaging and chemical analysis, the underlying mechanisms for the treatment-induced softening are identified and discussed. Based on the above results and analyses, major conclusions are drawn as follows:

1) The big data nanoindentation technique can be adopted as a practical, standalone toolkit to effectively and efficiently design and optimize hydraulic fracturing fluids for fracturing operations in different shale formations by screening chemical additives.
2) Both the DI water and KCl solution cause significant reduction in the mechanical properties of the shale, while some additives such as the Surf A used in this study, if properly selected, can prevent the mechanical softening. Therefore, rock-fluid interactions should be studied prior to the actual hydraulic fracturing operations.
3) Various softening mechanisms may contribute to the mechanical degradation caused by rock-fluid interactions. Among those are

pyrite oxidation, carbonate dissolution, clay hydration and swelling, and osmotic swelling.

CRediT authorship contribution statement

Shengmin Luo: Methodology, Software, Investigation, Validation, Writing - original draft, Data curation. **Yongkang Wu:** Software, Data curation. **Yucheng Li:** Software, Data curation. **Dongfang Wang:** Data curation, Writing - review & editing. **Daeyoung Kim:** Data curation, Writing - review & editing, Supervision. **Jinliang Song:** Software, Data curation, Formal analysis. **Guoping Zhang:** Conceptualization, Methodology, Resources, Writing - review & editing, Supervision, Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.coal.2021.103744.

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